

Investigation on Relationship between Common Mode Noise and Distribution of Parasitic Capacitance

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(Manuscript received July 27, 2022, revised Nov. 29, 2022)
J-STAGE Advance published date : Feb. 10, 2023

This study investigates the influence of parasitic capacitance on the common mode (CM) noise in a converter. A thermal pad on a switching device is necessary for sufficient heat dissipation. In addition, the thermal pad increases the CM capacitance between the printed circuit board and the ground plane. The parasitic capacitances form CM noise propagation paths. However, which node in the circuit and the corresponding parasitic capacitance influences the CM noise the most and the degree of the parasitic capacitance influence on the CM noise are unclear. In this study, the parasitic capacitance that influences the CM noise in a converter is identified, and the extent to which the parasitic capacitance affects the CM noise is investigated through simulations and experiments.

Keywords: EMI, common mode noise, parasitic capacitance

1. Introduction

Recently, with global concern for energy issue and environmental issue, electrified vehicles and electrified aircrafts have attracted attention as a kind of solutions to reduce energy consumption and gas emission. Renewable energy generation systems, such as photovoltaic and wind power generation, have also drawn attention as promising energy sources without gas emissions. Those systems include bidirectional DC–DC converters to manage the electric energy, and total consumption of electric energy in the world is growing up year by year. Thus, applications and markets of power converters are expanding, and a DC–DC converter is required to obtain high efficiency and high power density, in order to meet the growing demands.

Next generation of power semiconductors, such as gallium nitride (GaN), are considered to raise efficiency and power density of power converters. Such devices have characteristics of higher dv/dt ratio, which leads to lower switching loss and higher efficiency, than that of silicon devices. Low switching loss also enables high-frequency operation of power devices, which results in small passive components and high power density of a converter. Therefore, new generation power semiconductors are essential to power converters with high efficiency and high power density.

On the other hand, high dv/dt ratio of power devices is reported to cause large electromagnetic interference (EMI) in broad bandwidth^{(1)–(4)}. Therefore, certain amount of effort in perspective of electromagnetic compatibility (EMC) is necessary to take full advantage of the upcoming power semiconductors and to increase efficiency and power density of power converters.

Especially, conductive common-mode (CM) noise is reported as a problem. CM noise propagates through parasitic capacitances between thermal pads of devices and a ground plane. These capacitances inevitably exist because thermal pads are placed closely to a ground plane to achieve efficient heat dissipation. Many solutions to prevent CM noise have been proposed to reduce CM noise^{(4)–(11)}. The most common method to reduce CM noise is adding a CM noise filter that consists of CM choke coil and Y capacitor. Design methods of CM noise filter to cancel parasitic components in filters and exert full filtering performance have been studied in previous research^{(5)–(6)}. There are also CM noise reduction methods with noise current cancelation. The balanced circuit techniques realize CM noise cancelation^{(7)–(9)}, and CM noise can also be canceled by an additional winding to generate antiphase noise source^{(10)–(11)}. In the previous research^{(5)–(11)}, various original solutions, which are either increasing impedance in CM noise propagation paths or equivalently canceling parasitic components in the circuit, are proposed.

On the other hand, CM noise is mainly considered to propagate through parasitic capacitances between switching nodes and a ground plane^{(8)–(11)}, and the other parasitic capacitances to the ground plane is often ignored. Therefore, one of the most obvious solutions to decrease CM noise could be simply decreasing parasitic capacitance between switching nodes and a ground, while keeping same performance of switching and heat dissipation. However, it has been difficult to change parasitic capacitance to the ground without

This paper is based on Reference (15), which published in the International Power Electronics Conference, IPEC-Himeji 2022 - ECCE Asia (2022) ©2022 IEEJ.

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changing performance of switching and heat dissipation and therefore, it is difficult to investigate the influence of parasitic capacitance to the ground on CM noise through measurement. Thus, purpose of this paper is to introduce an implementation method to change parasitic capacitance corresponding to the switching node and investigate the influence of parasitic capacitance to the ground on CM noise through measurement.

This paper is the extended version of (15). As a new contents from (15), discussions on extraction of parasitic capacitance with finite element method (FEM), switching performance and the methods for arranging parasitic capacitance are added.

In the following chapters, first, circuit diagram of a DC–DC converter circuit with parasitic capacitance is indicated, and relationship between parasitic capacitance and CM noise is analyzed. Second, an implementation method of experimental circuits with different parasitic capacitance distribution is explained. Third, the CM noise spectrum of the DC–DC converter in simulation and measurement is shown in graphs. Then, methods to change distribution of parasitic capacitance to reduce CM noise are discussed. Finally, the whole discussion in this paper is concluded.

2. Parasitic Capacitance and CM Noise

2.1 Circuit Structure The circuit structure employed in the investigation is shown in Fig. 1. The circuit is a simple non-isolated DC–DC converter. V_{in} is the input voltage source. C_{in} and C_o are the input capacitor and the output capacitor respectively. L is the inductor. S_1 is the upper arm switching device and S_2 is the lower arm switching device. R_o is the resistive load. Line Impedance Stabilization Network (LISN) is placed between the input voltage source and the converter circuit to measure CM noise emitted from the converter circuit. The converter has a ground plane which is not shorted with any part of the main circuit of the converter.

In this paper, parasitic capacitance between a circuit and a ground is defined as CM capacitance. CM capacitance is distributed between the entire circuit and the ground plane. However, the circuit is divided into several nodes when illustrated in circuit diagram. Therefore, CM capacitance in a circuit diagram is expressed as if it is also distributed into each node. In Fig. 1, there are four CM capacitances: C_{vin} , C_h , C_{sw} , and C_1 . C_{vin} is the CM capacitance corresponding to the high voltage potential side of the input terminal. C_h , C_{sw} and C_1 are the CM capacitances corresponding to the high voltage potential side, the switching node, and the low voltage potential side of the half-bridge, respectively. Those CM capacitances are largely influenced by the positional relationships among patterns in the printed circuit board (PCB), switching devices, and the ground plane. Especially, C_h , C_{sw} and C_1 largely depend on the implementation of switching devices since those capacitances are connected directly to the switching devices.

2.2 CM Noise Analysis In this section, the relationship between the circuit and CM capacitance is analyzed mathematically. For CM noise analysis, the experimental circuit in Fig. 1 is converted into the CM noise equivalent circuit. The CM noise equivalent circuit is shown in Fig. 2. In Fig. 2, the input capacitance C_{in} and the output capacitance

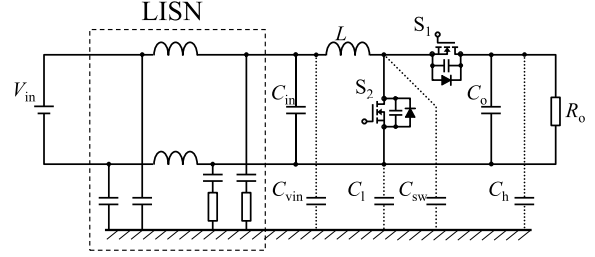


Fig. 1. Non-isolated boost DC–DC converter circuit

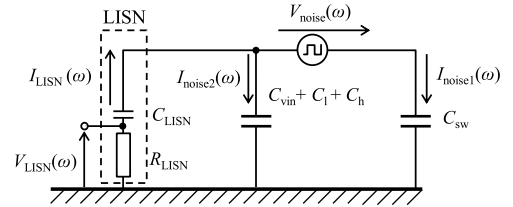


Fig. 2. CM noise equivalent circuit of Fig. 1

C_o is assumed to be short, and the inductor L is assumed to be open. The switching devices S_1 and S_2 are substituted with a noise voltage source $V_{noise}(\omega)$, where ω is noise angular frequency. LISN is expressed with a resistance R_{LISN} and capacitance C_{LISN} connected in series. $V_{LISN}(\omega)$ is the CM noise terminal voltage measured in the LISN. $I_{noise1}(\omega)$ is the current flowing through C_{sw} . $I_{noise2}(\omega)$ is the total current flowing through C_{vin} , C_h , and C_1 . $I_{LISN}(\omega)$ is the current flowing through LISN. The circuit equations of Fig. 2 is described as follows.

$$\begin{cases} V_{noise}(\omega) = \frac{I_{noise1}(\omega)}{j\omega C_{sw}} + \left(\frac{1}{j\omega C_{LISN}} + R_{LISN} \right) I_{LISN}(\omega) \\ 0 = \frac{I_{noise2}(\omega)}{j\omega (C_{vin} + C_h + C_1)} + \left(\frac{1}{j\omega C_{LISN}} + R_{LISN} \right) I_{LISN}(\omega) \\ V_{LISN}(\omega) = -R_{LISN} I_{LISN}(\omega) \\ I_{LISN}(\omega) = I_{noise1}(\omega) + I_{noise2}(\omega) \end{cases} \quad (1)$$

By solving the Eq. (1), CM noise voltage $V_{LISN}(\omega)$ in Fig. 1 is derived as follows.

$$V_{LISN}(\omega) = - \frac{j\omega C_{LISN} R_{LISN} V_{noise}(\omega)}{\left(1 + \frac{C_{LISN}}{C_{all}} \right) + j\omega C_{LISN} R_{LISN}} \frac{C_{sw}}{C_{all}} \quad (2)$$

Here, C_{all} indicates total CM capacitance and expressed as follows.

$$C_{all} = C_{vin} + C_h + C_{sw} + C_1 \quad (3)$$

To discuss difference in CM noise voltage between two different circuits, it is assumed that $V_{LISN,A}(\omega)$ is the CM noise voltage of circuit A where $C_{sw} = C_{sw,A}$, and $V_{LISN,B}(\omega)$ is the CM noise voltage of circuit B where $C_{sw} = C_{sw,B}$. From Eq. (2), $V_{LISN,A}(\omega)$ and $V_{LISN,B}(\omega)$ are expressed as in following equations.

$$\begin{cases} V_{\text{LISN}_A}(\omega) = -\frac{j\omega C_{\text{LISN}} R_{\text{LISN}} V_{\text{noise}}(\omega)}{\left(1 + \frac{C_{\text{LISN}}}{C_{\text{all}}}\right) + j\omega C_{\text{LISN}} R_{\text{LISN}}} \frac{C_{\text{sw}_A}}{C_{\text{all}}} \\ V_{\text{LISN}_B}(\omega) = -\frac{j\omega C_{\text{LISN}} R_{\text{LISN}} V_{\text{noise}}(\omega)}{\left(1 + \frac{C_{\text{LISN}}}{C_{\text{all}}}\right) + j\omega C_{\text{LISN}} R_{\text{LISN}}} \frac{C_{\text{sw}_B}}{C_{\text{all}}} \end{cases} \dots\dots\dots (4)$$

From Eq. (4), the relationship between $V_{\text{LISN}_A}(\omega)$ and $V_{\text{LISN}_B}(\omega)$ is expressed as in following equations.

$$\frac{V_{\text{LISN}_A}(\omega)}{V_{\text{LISN}_B}(\omega)} = \frac{C_{\text{sw}_A}}{C_{\text{sw}_B}} \dots\dots\dots (5)$$

Here, each of $V_{\text{noise}}(\omega)$ and C_{all} are assumed to be same in both circuits. From Eq. (5), CM noise voltage $V_{\text{LISN}}(\omega)$ changes depending on C_{sw} , even though $V_{\text{noise}}(\omega)$ and C_{all} are same in compared circuits. The relationships between CM noise voltage and CM capacitance in simulation and measurement are investigated in the following chapters.

3. Investigation Method

In the previous chapter, it is found that ratio of C_{sw} to C_{all} has large influence on CM noise terminal voltage. To compare CM noise, C_{vin} , C_{h} , C_{sw} , and C_1 should have different combinations of values among different circuits. To simplify the discussion, the noise voltage source $V_{\text{noise}}(\omega)$ and the total CM capacitance C_{all} should be kept same and then, CM noise changes depending only on change in C_{sw} . In this chapter, the switching devices and circuit design procedure of the experimental circuits are explained prior to the simulation and measurement.

3.1 Devices for Investigation In this paper, there are two kinds of switching devices employed to be S_1 or S_2 in the experimental circuits to change distribution of CM capacitance. Those are TP65H070LDG and TP65H070LSG, both of which are GaN FET with cascode device structure consisting of a high voltage GaN HEMT and a low voltage silicon MOSFET, and are manufactured by Transphorm Inc. It should be noted that these devices have the same electric characteristics including voltage and current ratings, parasitic capacitances among terminals, rise time and fall time and so on, but different footprints. The terminal assignments on the bottom surfaces of TP65H070LDG and TP65H070LSG are shown in Fig. 3(a) and (b). As shown in Fig. 3, each device has one large thermal pad and the other two small terminals on the bottom surface. The thermal pads have the same area,

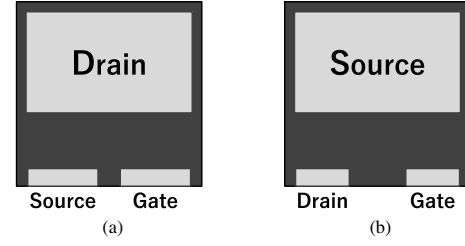


Fig. 3. Bottom view of (a) TP65H070LDG and (b) TP65H070LSG

but the other pairs of terminals are slightly different in size, between the two devices. The most obvious difference between two devices is that the thermal pads are connected to different terminals. In TP65H070LDG shown in Fig. 3(a), the drain terminal is connected to the thermal pad, and the source terminal and the gate terminal are connected to the other two small terminals. On the other hand, in TP65H070LSG shown in Fig. 3(b), the source terminal is connected to the thermal pad, and the drain terminal and the gate terminal are connected to the other two small terminals. With the different footprints and the same electric and thermal characteristics, distribution of CM capacitance should be different between the two devices while the noise voltage source $V_{\text{noise}}(\omega)$ is kept same.

To examine the CM capacitance in detail, the cross-section of a mounted device is shown in Fig. 4. Double-sided PCBs are employed for the experimental circuits. A MOSFET is soldered on the top surface of the PCB, and an insulation sheet and a heat sink are fixed on the other surface of the PCB to lower the thermal resistance. The thermal pad on the MOSFET is electrically and thermally connected to the thermal pattern on the bottom surface through thermal vias for efficient heat dissipation. The thermal pattern on the bottom surface is designed large enough and placed close enough to the heat sink, and the heat sink is connected to the ground. Therefore, a thermal pad in each device inevitably has large CM capacitance. This CM capacitance is defined as C_{thermal} in this paper. On the other hand, a drain terminal or source terminal assigned to a small terminal, which are indicated as the bottom left terminals in Fig. 3(a) and (b), also have corresponding CM capacitance. But unlike C_{thermal} , this capacitance can be minimized with PCB pattern design, since the small terminal is not necessarily connected to the bottom surface for heat dissipation. This CM capacitance is defined as C_{small} . Thus, if C_{small} is minimized to decrease CM noise,

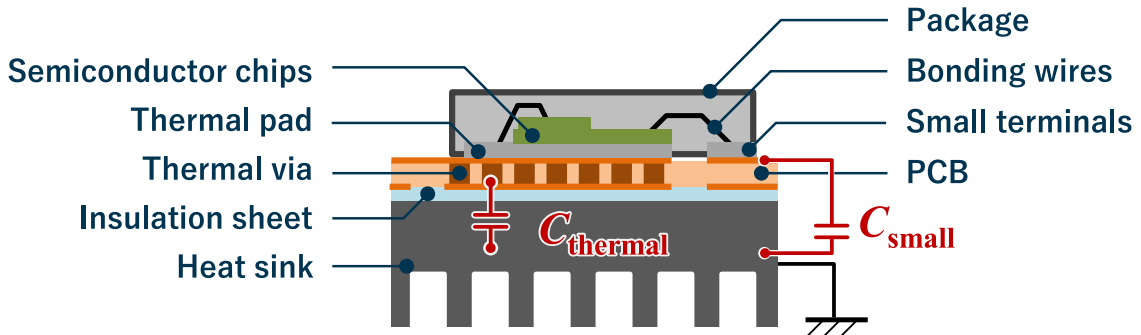


Fig. 4. Cross section of a switching device, a PCB, an insulation sheet, and a heat sink

Table 1. Parasitic capacitance and combinations of FETs

		Case 1	Case 2	Case 3	Case 4
Circuit diagram of a half-bridge *□: a thermal pad					
Thermal pad	S ₁	Drain	Drain	Source	Source
	S ₂	Drain	Source	Drain	Source
C_{sw}		$C_{thermal} + C_{small}$	$2C_{small}$	$2C_{thermal}$	$C_{thermal} + C_{small}$
$C_{vin} + C_h + C_1$		$C_{thermal} + C_{small} + C_{other}$	$2C_{thermal} + C_{other}$	$2C_{small} + C_{other}$	$C_{thermal} + C_{small} + C_{other}$
C_{all}		$2C_{thermal} + 2C_{small} + C_{other}$			

$C_{thermal}$ naturally becomes much larger than C_{small} . In this paper, the CM capacitances corresponding to the gate terminals are considered small enough and ignored for simplification. The other CM capacitances except $C_{thermal}$ and C_{small} of S_1 and S_2 is defined as C_{other} , which is assumed to be distributed into C_{vin} , C_h , and C_1 in Fig. 1. $C_{thermal}$ and C_{small} are shown in Fig. 4 as well. Each of $C_{thermal}$, C_{small} and C_{other} differs depending on the design of circuit pattern. $C_{thermal}$ and C_{small} of switching devices S_1 and S_2 and C_{other} can be distributed into C_{vin} , C_h , C_{sw} , and C_1 . The relationship among those CM capacitances is expressed as in following equation.

$$C_{all} = C_{vin} + C_h + C_{sw} + C_1 = 2C_{thermal} + 2C_{small} + C_{other} \quad (6)$$

The relationship between combinations of switching devices and C_{sw} is explained as follows. There are two switching devices S_1 and S_2 in the converter in Fig. 1, and two sorts of FETs to be employed in the experimental circuits: TP65H070LDG and TP65H070LSG. Thus, there are four possible combinations of FETs. Circuit diagram of half-bridge in all combinations are shown in Table 1. In Table 1, a blue square indicates a thermal pad, and red letters indicate switching nodes. In addition, distribution of CM capacitance $C_{thermal}$, C_{small} and C_{other} in all combinations are also shown in Table 1. C_{other} is included in $C_{vin} + C_h + C_1$ in all cases. $C_{thermal}$ should be same among all cases for equal heat dissipation. In case 2 in Table 1, neither of the thermal pads is connected to the switching node. Therefore, $C_{thermal}$ is not included in C_{sw} and C_{sw} is the smallest in case 2 among all cases. On the other hand, in case 3 both of thermal pads are connected to the switching node. Therefore, C_{sw} is the largest in case 3 among all cases. In case 1 and 4, one of the thermal pads is connected to the switching node and C_{sw} is same between case 1 and 4. In the following section, circuit implementation procedure to achieve above distributions of CM capacitance is discussed.

3.2 Circuit Implementation For precise comparison, the CM capacitances should be designed carefully. In this section, several procedures of circuit implementation in the experimental circuits are explained.

First, $C_{thermal}$ should be designed same for equal heat dissipation among all experimental circuits. Therefore,

experimental circuits should have same $C_{thermal}$ but different C_{sw} . The cross section of an experimental circuit is shown in Fig. 5(a). Each of MOSFETs S_1 and S_2 is mounted on the top surface of PCB and therefore, S_1 and S_2 should be cooled through thermal patterns on the bottom surface. The patterns on the bottom surface of PCB in all cases are shown in Fig. 5(b) to (e). All bottom surfaces have same form and size, but different connection because of different selection of MOSFETs. Bottom surface of each PCB is divided in 3 patterns. Two of those patterns are thermal patterns connected to thermal pads of S_1 and S_2 as shown in Fig. 5(b) to (e). These thermal patterns are designed to have same area so that heat dissipation performance and $C_{thermal}$ of each device becomes same. A thermal pattern is indicated with red line when the pattern is connected to switching node. The other pattern occupying large part of bottom surface is connected to source terminal of S_2 which corresponding to C_1 and because of the size of this circuit pattern C_1 should be dominant in C_{all} . From Fig. 5, neither of thermal patterns is connected to the switching node in the case 2. Therefore, the C_{sw} is the smallest in case 2 among all cases as described in the previous section. On the other hand, both of thermal patterns are connected to the switching node in case 3. Therefore, the C_{sw} is the largest in case 3 among all cases as also described in the previous section. Here, CM noise should depend only on C_{sw} since all bottom surfaces have same form and size and values of C_{all} are close enough in all PCBs.

As assembling an experimental circuit, a PCB, an insulation sheet, and a heat sink are fixed together with screws. Here, polycarbonate screws are employed as fixing tools instead of metal screws to avoid additional EMI unrelated to the PCB design. The screws in the experimental circuits are shown in Fig. 6. There are three holes for screws on each PCB. All screws are tightened with same torque by using a torque driver, so that the distance between the PCB and heat sink in each experimental circuit becomes same. Here, it should be noted that the following simulation and measurement results of CM noise differ only because of the difference in distribution of CM capacitance, since FETs with the same electric characteristics are employed for same $V_{noise}(\omega)$, and total CM capacitances C_{all} is designed to be same among all experimental circuits, as explained so far.

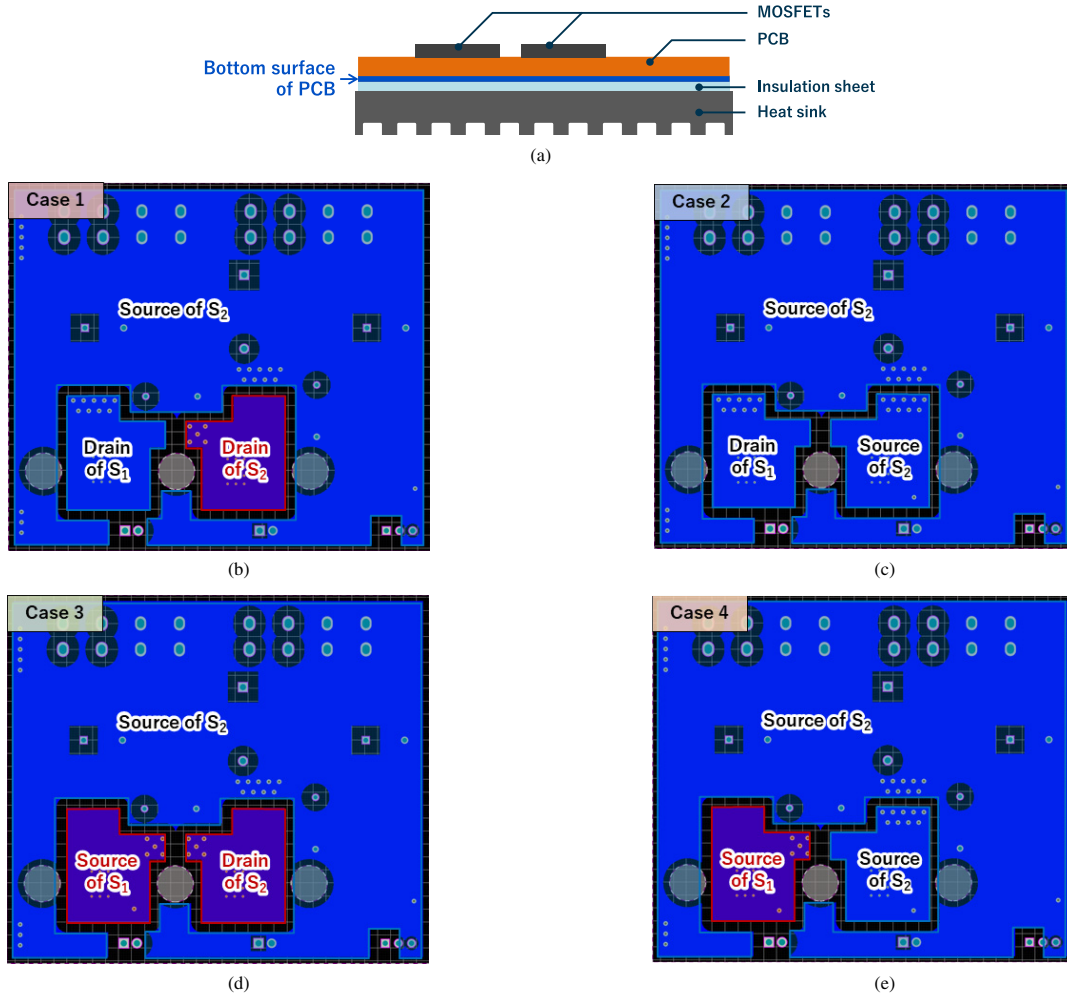


Fig. 5. (a) Cross section of an experimental circuit, and bottom circuit patterns of experimental circuits with different device combinations: (b) case 1, (c) case 2, (d) case 3, and (e) case 4 in Table 1

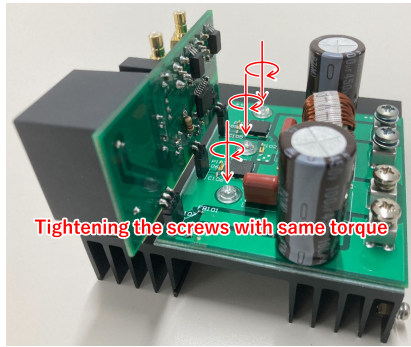


Fig. 6. Polycarbonate screws in an experimental circuit

4. Simulation

In this chapter, the CM noise terminal voltage in each case of device combination is compared in simulation. First, the CM capacitance in each experimental circuit is extracted through FEM simulation. Then, CM noise terminal voltage is analyzed through circuit simulation.

4.1 CM Capacitance Extraction For circuit simulation, CM capacitance should be set to the same values as those in the experimental circuit. However, it is difficult to measure each of C_{vin} , C_h , C_{sw} and C_l separately in the

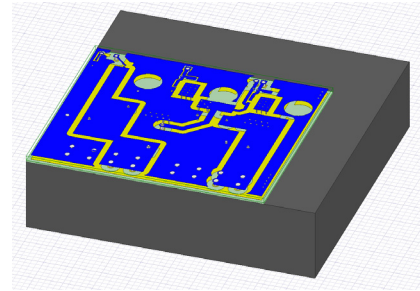


Fig. 7. 3D model of the experimental circuit for case 1

experimental circuit, even though the total CM capacitance C_{all} is measurable. Therefore, ANSYS Q3D is employed to extract each of the CM capacitances from the experimental circuit models. The 3D model of an experimental circuit is shown in Fig. 7 as an example. Each 3D model consists of a PCB, insulation sheet and a heat sink. The thickness of each insulation sheet is designed so that C_{all} 's in simulation and measurement become same values. The extracted CM capacitance values are shown in Fig. 8. In Fig. 8, C_{sw} shows close values in case 1 and case 4, as indicated in Table 1. Compared to case 1 and case 4, C_{sw} is much smaller in case 2, and much larger in case 3, as explained in Chapter 3. The total CM capacitance C_{all} shows similar value among all cases, as

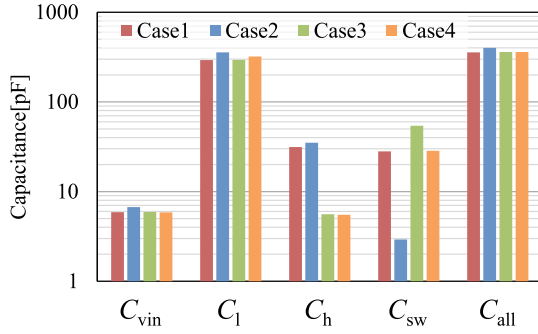
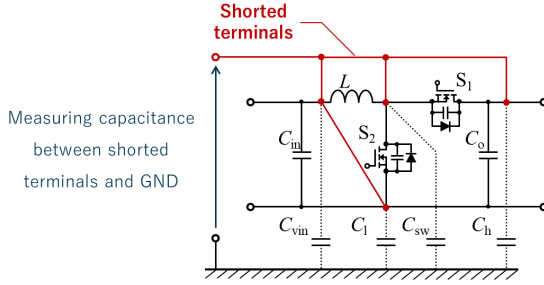


Fig. 8. CM capacitances extracted by ANSYS Q3D

Fig. 9. Circuit connection to measure C_{all} Table 2. Comparison of total CM capacitance C_{all} in the simulation and measured

		Case 1	Case 2	Case 3	Case 4
C_{all}	Simulation	358	401	360	360
[pF]	Measurement	397	355	394	396
	Error [%]	-9.8	13.2	-8.53	-9.11

designed. C_1 is dominant in C_{all} among all cases since large part of bottom surface pattern is connected to source terminal of S_2 , which corresponds to the CM capacitance C_1 . To verify the consistency of extracted capacitance values, C_{all} of all experimental circuits is also measured in advance. The circuit connection to measure C_{all} is shown in Fig. 9. To measure C_{all} , all nodes in an experimental circuit is shorted and the capacitance between shorted nodes and the ground is measured as C_{all} with an impedance analyzer E4990A manufactured by Keysight Technologies. The comparison of C_{all} values in simulation and measurement is shown in Table 2. As shown in Table 2, the errors of total capacitance extracted are within 13.2% in all cases. These capacitance values are used in the following circuit simulation.

4.2 CM Noise Analysis CM noise terminal voltage is compared in the circuit simulation among all cases shown in Table 1. The parameters for investigation are shown in Table 3. In the simulation, $V_{noise}(\omega)$ corresponds to drain-source voltage of ideal MOSFET and does not change among all cases. Each of C_{vin} and C_1 in the simulation circuit includes additional capacitance by 155 pF, which originates from the CM capacitance of input cable between LISN and the PCB. The simulation results are shown in Fig. 10. As described in Chapter 3, cases 1 and 4 show almost the same CM noise terminal voltage, which are about 6 dB below the result of the case 3 in broad bandwidth. Case 2 shows the smallest CM noise terminal voltage which is about 26 dB below the result of case 3. These differences among simulation results correspond to the difference among C_{sw} in all cases shown in

Table 3. Parameters for investigation

Parameter	Value
Input voltage	160 V
Output voltage	240 V
Input power	524 W
Switching frequency	100 kHz
Duty ratio	0.333
R_{LISN}	25 Ω
C_{LISN}	0.2 μ F

Table 4. Measurement equipment

Instrument	Part number	Manufacturer
LISN	NNBM 8124	SCHWARZBECK
CM/DM switch	CMDM 8700 B	SCHWARZBECK
Spectrum analyzer	RSA306B	Tektronix

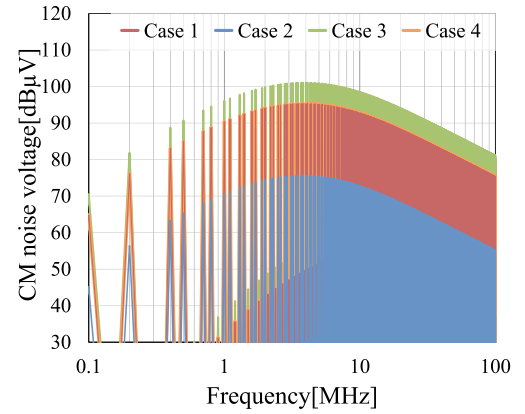


Fig. 10. CM noise terminal voltage in simulation result

Table 1 and Fig. 8, where total CM capacitance C_{all} and noise voltage source $V_{noise}(\omega)$ do not change in all cases. Therefore, simulation results show that CM noise terminal voltage largely depends on the value of C_{sw} even though the total CM capacitance C_{all} and noise voltage source $V_{noise}(\omega)$ are same among all cases.

5. Measurement

The CM noise terminal voltage in each case of device combination is compared in measurement as well. The experimental parameter also refers to Table 3. The measurement equipment is shown in Table 4. The measurement results are shown in Fig. 11. Here, the CM/DM switch CMDM 8700 B has 20 dB $\pm$ 3 dB attenuation as extracting CM noise. Therefore, the measurement results shown in Fig. 11 have been corrected as much as attenuated by CMDM 8700 B. Same as the simulation result, case 3 shows the largest CM noise terminal voltage and case 2 shows the smallest CM noise terminal voltage in broad bandwidth. Between case 2 and case 3, there is almost 20 dB difference in the frequency under 20 MHz. Case 1 and 4 shows almost same CM noise level under 20 MHz which is also about 4 dB to 6 dB below that of case 3. Relative difference among the spectra in measurement results is very similar to that of the spectra in simulation results under 30 MHz.

In the frequency band over 30 MHz, the spectra in measurement results decrease more steeply than those in simulation data do. This difference in spectra originates from the difference in noise voltage source. The spectra of drain-

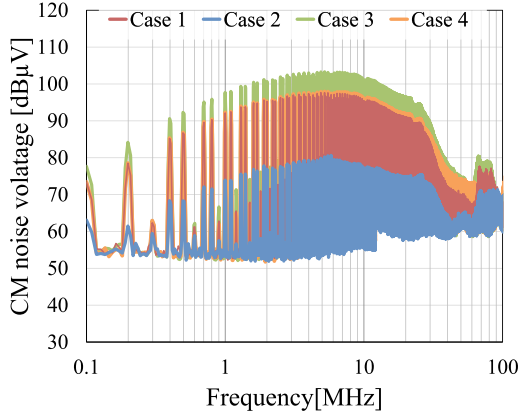
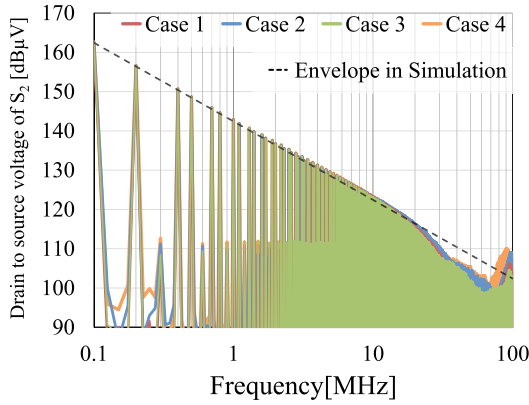


Fig. 11. CM noise terminal voltage in measurement result

Fig. 12. Spectra of drain-source voltage of S_2 in the measurement compared with that in the simulation

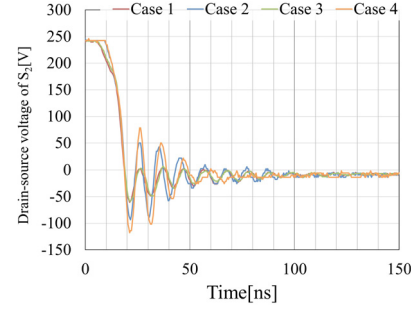
source voltage, which corresponds to $V_{\text{noise}}(\omega)$, of each case in measurement are shown and compared with that in the simulation in Fig. 12. In Fig. 12, spectra in the measurement also decrease more steeply than that in the simulation. Therefore, from Fig. 12 and Eq. (2), CM noise voltage $V_{\text{LISN}}(\omega)$ in measurement over 30 MHz steeply decreases depending on frequency characteristic of $V_{\text{noise}}(\omega)$.

From Fig. 11, the measurement results show very similar characteristics to the simulation results under 30 MHz. Thus, it is verified also in measurement that the distribution of CM capacitance has a large influence on the CM noise.

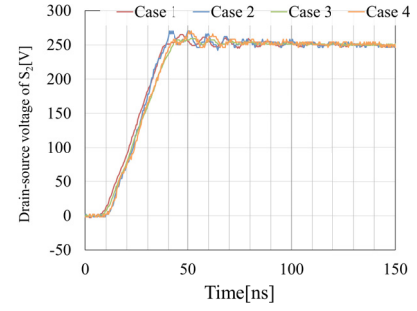
In addition, switching performance of each experimental circuit is evaluated. The switching waveforms are shown in Fig. 13 and efficiency of all the circuits are shown in Table 5. From Fig. 13, the waveforms have very close dv/dt characteristics in all cases. However, the resonance should not degrade CM noise reduction performance other than resonant frequency which in this case is around 80 MHz to 100 MHz. On the other hand, from Table 5, the conversion efficiency of all experimental circuits shows similar values. From Fig. 13 and Table 5, CM noise voltage among all cases differ without difference in switching performance of devices.

6. Discussion

From simulation and measurement, it is verified that CM noise decreases when C_{sw} , which is CM capacitance corresponding to switching node, decreases, even though the total



(a)



(b)

Fig. 13. Drain to source voltage waveforms of S_2 at (a) turn-on and (b) turn-off

Table 5. Efficiency of experimental circuits

	Case 1	Case 2	Case 3	Case 4
Efficiency [%]	97.8	97.8	97.9	97.9

CM capacitance and noise voltage source do not change. In short, the distribution of CM capacitance has large influence on CM noise. As next steps, several approaches to connect the investigation results in this paper to actual CM noise solutions are discussed in this chapter.

One easy method to control CM capacitance is to employ devices with selectable thermal pads, just like TP65H070LDG and TP65H070LSG discussed in this paper. For example, GS66508P, which is a GaN transistor manufactured by GaN Systems Inc., has thermal pad floating from both drain and source terminal. The bottom surface and circuit symbol of GS66508P are shown in Fig. 14. The thermal pad is internally connected to the substrate and should be externally connected to either the source or drain terminal. In (12), operation of GS66508P in both case of thermal pad connection is investigated. Therefore, GS66508P should reduce CM noise by connecting thermal pad to reduce CM capacitance corresponding to switching node.

In terms of device structure, there is a half bridge module proposed in (13). The proposed module in (13) is shown in Fig. 15. This module has a thermal pad connected to source terminal of lower arm and should have small CM capacitance corresponding to the switching node.

On the other hand, in terms of circuit configuration, reversed configuration is proposed in (14). A conventional DC-DC converter and the reversed circuit configuration are shown in Fig. 16. In this configuration, source terminal of MOSFET, which often is not assigned to thermal pad in a vertical MOSFET, is connected to switching node. Therefore, with the reversed configuration, a converter can reduce CM capacitance corresponding to switching node and reduce CM

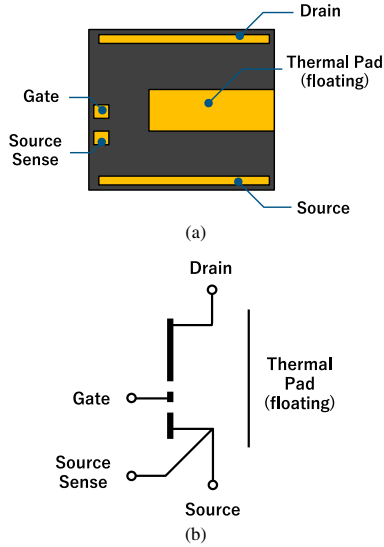


Fig. 14. GS66508P discussed in (12) in (a) bottom surface of package and (b) circuit symbol

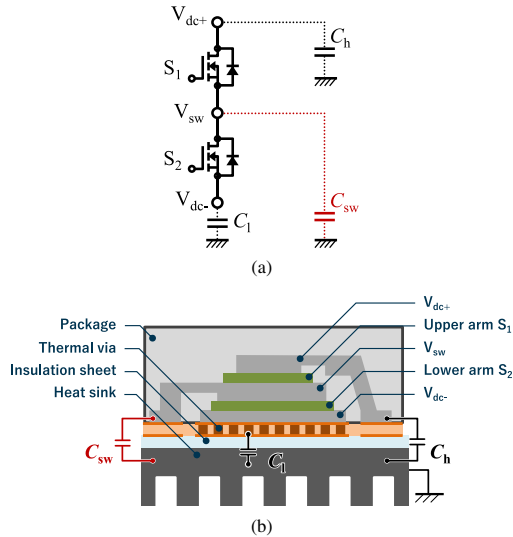


Fig. 15. Half-bridge module proposed in (13) in (a) circuit diagram and (b) cross section

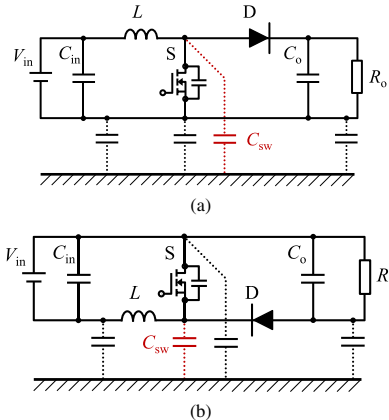


Fig. 16. (a) Conventional DC-DC converter and (b) reversed circuit configuration proposed in (14)

noise without changing thermal pad.

7. Conclusion

In this paper, the relationship between CM noise and dis-

tribution of CM capacitance is clarified.

First, the CM noise in a non-isolated DC-DC converter is analyzed, and the CM capacitance corresponding to switching node C_{sw} is found to have large influence on CM noise. Second, the circuit design and implementation necessary for the proper investigation is explained. Third, the relationship between CM noise and the distribution of the CM capacitance is investigated and verified based on simulation and measurement. Lastly, several approaches to control CM capacitance are discussed. As a conclusion, distribution of the CM capacitance is verified to have large influence on CM noise, and therefore, CM noise can be decrease in large scale only by arranging distribution of CM capacitance.

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